

Incident Report – Phishing Email & Malicious Macro Execution

Title:

Phishing Email Investigation – Excel 4.0 Macro Malware Execution

Scenario:

A phishing email containing a malicious Excel 4.0 macro attachment was delivered to a user endpoint. The attachment was executed, resulting in outbound communication with malicious Command & Control (C2) infrastructure. A SOC investigation was initiated to determine the extent of compromise.

Tools used:

- SIEM / Log Management Platform
- Endpoint Detection & Response (EDR)
- Sandbox Analysis Environment
- Windows CMD History Review
- Browser History Investigation
- Network Connection Monitoring

Alert / initial evidence:

SOC alert rule 'SOC146 - Phishing Mail Detected - Excel 4.0 Macros' triggered after detection of suspicious activity related to an email attachment. Sandbox analysis identified malicious behavior and C2 communication attempts.

Investigation steps:

1. Reviewed SIEM alert details and timestamps
2. Performed sandbox analysis on the Excel attachment
3. Identified C2 addresses generated by the malware
4. Searched log management platform for communication with malicious addresses
5. Investigated LarsPRD endpoint activity
6. Reviewed browser history and network connections
7. Examined CMD history for suspicious commands
8. Confirmed execution of regsvr32 associated with Excel 4.0 macros

Findings:

The malicious Excel attachment successfully executed on the LarsPRD endpoint. Sandbox analysis confirmed malicious behavior and generated multiple C2 addresses. Endpoint investigation identified suspicious outbound communication and execution of regsvr32 commands. The incident was classified as a TRUE POSITIVE phishing attack.

Indicators of Compromise:

- Malicious Excel 4.0 Macro attachment
- regsvr32 execution detected in CMD history
- Outbound communication to malicious C2 infrastructure
- Suspicious browser and network activity
- Affected device: LarsPRD

MITRE ATT&CK mapping:

- T1566.001 – Phishing: Spearphishing Attachment
- T1204 – User Execution
- T1059.001 – PowerShell / Command Execution
- T1218.010 – regsvr32
- T1071 – Application Layer Protocol

Impact assessment:

The endpoint was compromised following execution of the malicious attachment. The malware established outbound communication with external malicious infrastructure, creating risk of further payload delivery, credential theft, or persistence.

Recommended actions:

- Immediately isolate the LarsPRD endpoint
- Block malicious IP addresses and domains
- Reset potentially compromised user credentials
- Conduct full malware scan and forensic investigation
- Disable or restrict Office macro execution policies
- Improve phishing awareness training for users
- Increase monitoring for similar phishing activity

Conclusion:

The phishing investigation confirmed successful execution of a malicious Excel 4.0 macro attachment. Rapid detection through SIEM alerts, combined with sandbox and endpoint analysis, enabled timely identification of the compromise and response actions.